



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

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Stack using Array

STACK USING ARRAY

"A web browser maintains a history of recently visited pages. Write a C program to implement a stack using an array and perform Push, Pop, Peek, and Display operations while handling overflow and underflow conditions.

Test Case 1

Input

Stack Size = 5

Push: 10, 20, 30

Display

Output

Stack Elements:

30

20

10"

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int stack[MAX];
6 int top = -1;
7 int size;
8
9 void push(int value) {
10     if (top == size - 1) {
11         printf("Stack Overflow\n");
12     } else {
13         stack[++top] = value;
14     }
15 }
16
17 int pop() {
18     if (top == -1) {
19         printf("Stack Underflow\n");
20         return -1;
21     } else {
```

OUTPUT

```
Stack Elements:
30
20
10
```

INPUT

5
10 20 30

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